ARC-AGI-3 Failure Mode Analysis

and the TTDB / Locus Framework Response

Toot Toot Engineering · Locus Project · May 2026 · CC0-1.0

Abstract

ARC Prize Foundation's May 2026 analysis of GPT-5.5 and Opus 4.7 on ARC-AGI-3 identified three structural failure modes in frontier large-language-model (LLM) agents: failure to build a global world model from local observations, inappropriate analogical mapping from training data, and an inability to carry reward signal across levels. This document examines how the TTDB (Toot Toot Database) knowledge-graph architecture, as implemented in the Locus framework, directly addresses each failure mode at a structural level, with the *companion_arcprize.md* dataset serving as a concrete demonstration vehicle for the approach.

1. Background and Sources

1.1 ARC-AGI-3

ARC-AGI-3 is a benchmark suite of 135 hand-crafted novel environments designed to test abstract reasoning without relying on cultural or domain knowledge. Agents receive no instructions and must infer rules from sparse feedback, form and revise hypotheses, and transfer learning across levels. GPT-5.5 scored 0.43% and Opus 4.7 scored 0.18% on the semi-private dataset — very low scores that nonetheless reveal rich qualitative differences in reasoning strategy.

1.2 The Locus / TTDB Framework

TTDB (MyMentalPalaceDB) is an offline-first, deterministic semantic knowledge-graph format developed under the Locus framework by Toot Toot Engineering (GitHub: toot-toot-engineering). It encodes knowledge as a graph of coordinate-anchored percept nodes connected by typed, weighted edges (deltas). Its philosophical grounding is Jakob von Uexküll's umwelt theory: every node represents a perception relative to an agent's position in a conceptual or physical space, and all meaning emerges from the edges between nodes — the delta being the primary datum. The *companion_arcprize.md* dataset maps ARC-AGI-3 concepts into this graph structure, anchored at coordinate lat60lon20 in Locus's addressing scheme.

2. The Three ARC-AGI-3 Failure Modes

The ARC Prize analysis team reviewed 160 game replays and reasoning traces, identifying three dominant failure patterns common to both models.

FM-1: True Local Effect, False World Model

Models correctly perceived local action effects ("ACTION3 rotates this object") but consistently failed to synthesize these into a global causal model ("rotation controls which face receives a value, so orient before dipping"). Observations accumulated without anchoring in a coherent theory of the environment.

FM-2: Wrong Level of Abstraction from Training Data

Models mapped unfamiliar mechanics onto known games — Tetris, Frogger, Sokoban, Breakout — because surface visual similarity triggered a training-data analogy. The analogy then hijacked action selection, causing the model to test the wrong affordances for dozens of steps before (sometimes) correcting.

FM-3: Solved the Level, Didn't Learn the Game

Even a successful level completion did not guarantee a correct world model. Opus 4.7 solved ka59 Level 1 in 37 actions using a wrong primitive (teleportation instead of shape-matching), then carried that misconception forward. Early success masked a missing or distorted causal understanding, and the partial win became a confident scaffold for an incorrect Level 2 strategy.

Model-Level Contrast

The analysis further distinguished the failure modes by model: Opus 4.7 exhibited **wrong compression** — rapid but incorrect theory formation followed by aggressive execution. GPT-5.5 exhibited **failure to compress** — broad hypothesis generation without commitment, drifting through multiple genre analogies without settling on an action plan.

Hypothesis formation	Fast, confident	Wide, uncommitted
World-model accuracy	Often wrong	Often correct but unused
Action execution	Aggressive on wrong theory	Hesitant; re-opens genre space
Failure label	Wrong compression	Failure to compress

3. TTDB Architectural Response

Each failure mode maps onto a structural property of TTDB. The following sections trace these correspondences precisely.

3.1 Addressing FM-1: The Delta as Primary Datum

The root cause of FM-1 is that LLMs accumulate observations as sequential tokens without a persistent relational structure that enforces causal closure. TTDB's architecture inverts this: the *edge* (delta) between percept nodes is the primary datum, not the node itself. Every observation is stored as a typed, directed, weighted relationship between two coordinate-anchored entities.

In an ARC-AGI-3 context, this means that when an agent observes "ACTION3 rotated object X," TTDB does not merely append this as text. It encodes:

@PERCEPT:ACTION3 --[CAUSES:ROTATION]--> @PERCEPT:OBJECT_X @PERCEPT:OBJECT_X --[ORIENTATION:CONTROLS]--> @PERCEPT:FILL_FACE

The global world model is not inferred post-hoc — it is the graph itself. Any agent reading the TTDB for this environment already has the causal chain encoded as explicit edges. FM-1 cannot occur because local effects are structurally forced to connect to global consequences at write time.

3.2 Addressing FM-2: Coordinate Anchoring Prevents Spurious Analogy

FM-2 arises because LLMs index knowledge by semantic similarity: a grid with falling objects activates "Tetris" or "Powder Toy" schema. TTDB anchors every node at an explicit coordinate address (e.g., lat60lon20 in the Locus addressing scheme) that is independent of semantic content. Retrieval is positional, not similarity-based.

When an agent navigates TTDB for an ARC-AGI-3 environment, it retrieves nodes by their address in the knowledge graph — the specific percept nodes for *this* environment's mechanics — rather than activating a cosine-similar training memory. The umwelt principle enforces this: the agent's knowledge is defined by what it has *perceived at this location*, not by what it has seen elsewhere that looks similar. False analogies require a background context that TTDB structurally withholds.

3.3 Addressing FM-3: Explicit Confidence Edges and Level Boundaries

FM-3 occurs because there is no persistent mechanism to distinguish "I won Level 1" from "I *understood* Level 1." TTDB edges carry typed metadata including confidence weights and provenance tags. An agent writing to TTDB after Level 1 can encode the win as:

@PERCEPT:LEVEL1_OUTCOME --[CONFIDENCE:LOW, CAUSE:UNKNOWN]--> @PERCEPT:WIN_STATE @PERCEPT:LEVEL1_MECHANIC --[HYPOTHESIS:UNVERIFIED]--> @PERCEPT:TELEPORT_THEORY

The low-confidence and unverified tags are then available when the agent enters Level 2. Rather than inheriting a false scaffold, it inherits a structured uncertainty marker. The TTDB graph for Level 2 starts with explicit epistemic state from Level 1, which an agent loop (Agent 32 in the Locus stack) can interrogate before committing to a strategy. This is continual learning encoded structurally rather than hoped for behaviorally.

4. The companion_arcprize.md Dataset

The *companion_arcprize.md* file, served at the Locus demo endpoint (antfriend.github.io, toot=lat60lon20), is a TTDB knowledge graph whose node topology encodes the ARC Prize problem space in umwelt terms. Its coordinate anchor at lat60lon20 locates it in Locus's abstract semantic geography. The file serves two functions:

• **Demonstration:** It shows how ARC-AGI-3 concepts — environments, mechanics, failure modes, hypotheses — can be represented as a graph of percept-delta pairs rather than as prose or token sequences.

• **Scaffolding:** A Locus agent (Agent 32 on ESP32) or any consumer of the TTDB format can load this graph and immediately access a structured causal map of the ARC Prize problem space, including typed edges for action-effect relationships, confidence annotations, and level-transition boundaries.

The companion dataset is CC0-licensed and designed to be extended: researchers can add new environment nodes, annotate failure modes from additional model runs, and build cross-environment causal chains — all within the TTDB schema, without custom tooling.

5. Correspondence Summary

ARC-AGI-3 Failure Mode	Root Cause	TTDB / Locus Mechanism
FM-1: Local effect, no world model	Observations not structurally connected	Delta (edge) as primary datum; causal closure enforced at write time
FM-2: Spurious training-data analogy	Retrieval by semantic similarity	Coordinate-anchored retrieval; umwelt isolation per environment
FM-3: Win without understanding	No persistent epistemic state across levels	Confidence-weighted and provenance-tagged edges; structured uncertainty propagation
Cross-model: compression asymmetry	No explicit commitment mechanism	Agent 32 loop: hypothesis nodes promote to world-model nodes when evidence threshold met

6. Limitations and Open Questions

• TTDB addresses the *representational* failure modes. It does not, by itself, provide a perception front-end capable of reading ARC-AGI-3 pixel frames into the graph. A vision-to-TTDB encoder would be required for a complete agent.

• The confidence-threshold mechanism for promoting hypotheses to world-model facts (FM-3 response) is described architecturally but not yet formally specified in a Locus RFC. This is an open engineering task.

• The @PERCEPT namespace formalization (vs. the proposed @TTP: prefix) remains an open question in the Locus RFC series and would affect the canonical encoding of environment nodes in companion datasets.

• ARC-AGI-3 environments require real-time sequential action; TTDB as currently specified is a storage/reasoning layer, not a real-time action loop. Agent 32's hardware abstraction layer (A32-RFC-0003) would need to be extended for interactive game environments.

7. Conclusions

The ARC-AGI-3 analysis reveals that frontier LLM agents fail not because they lack knowledge, but because their knowledge lacks *structure*. Observations do not automatically become causal models. Training memories intrude on novel environments. Success metrics do not propagate epistemic uncertainty.

TTDB's umwelt-grounded, delta-primary architecture addresses each of these structural deficits. By making causal edges explicit, isolating environments by coordinate address, and encoding confidence as a first-class graph property, TTDB provides the representational substrate that ARC-AGI-3 exposes as missing from current LLM approaches.

The *companion_arcprize.md* dataset operationalizes this argument as a navigable knowledge graph, available at the Locus demo and under CC0 for research use. Further work should address the perception front-end, formal RFC specification of confidence propagation, and integration with the Agent 32 real-time action loop.

References

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